

Secured Azure Web App Integration with MySQL on Linux VM

Objective

To deploy a PHP-based web application in Azure App Service connected securely to a MySQL database hosted on a Linux Virtual Machine through VNet integration, ensuring full governance, monitoring, and backup configurations.

Tools & Technologies Used

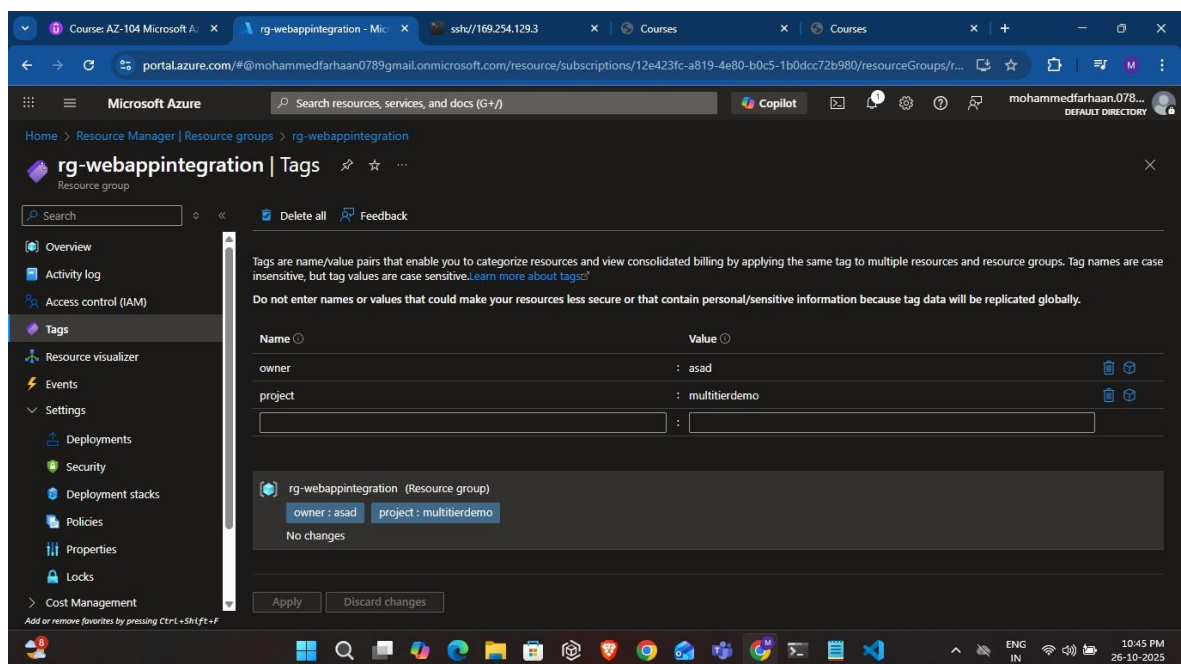
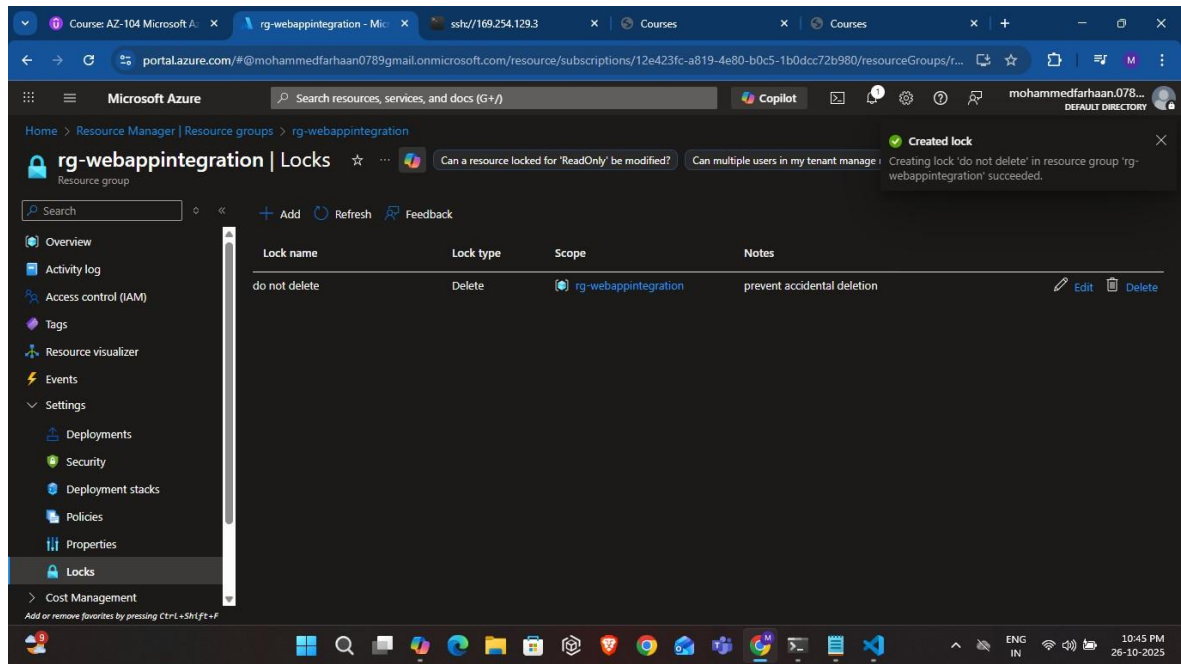
- 1 Microsoft Azure Portal
- 2 Azure Virtual Network (VNet)
- 3 Azure App Service (Linux Plan)
- 4 Linux VM (Ubuntu) with MySQL Server
- 5 Azure Recovery Services Vault (RSV)
- 6 Azure Monitor & Log Analytics Workspace
- 7 Azure Alerts & Action Groups & NSG, Tags, Locks, and RBAC Roles

AZ-104 Domains Covered

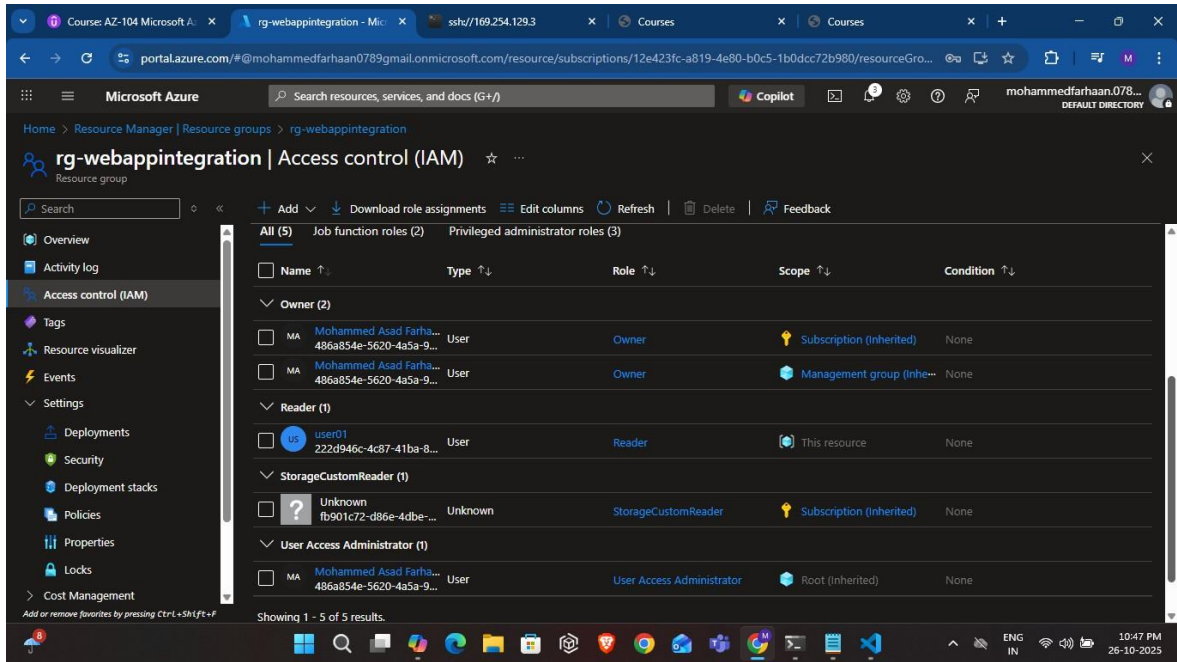
- 1 Manage Azure identities and governance (RBAC, Locks, Tags)
- 2 Implement and manage storage & compute resources (VM, Disks, Backup)
- 3 Configure and manage virtual networking (VNet, Subnets, NSG)
- 4 Manage App Services (Deployment, Monitoring, Alerts)

Deployment Steps

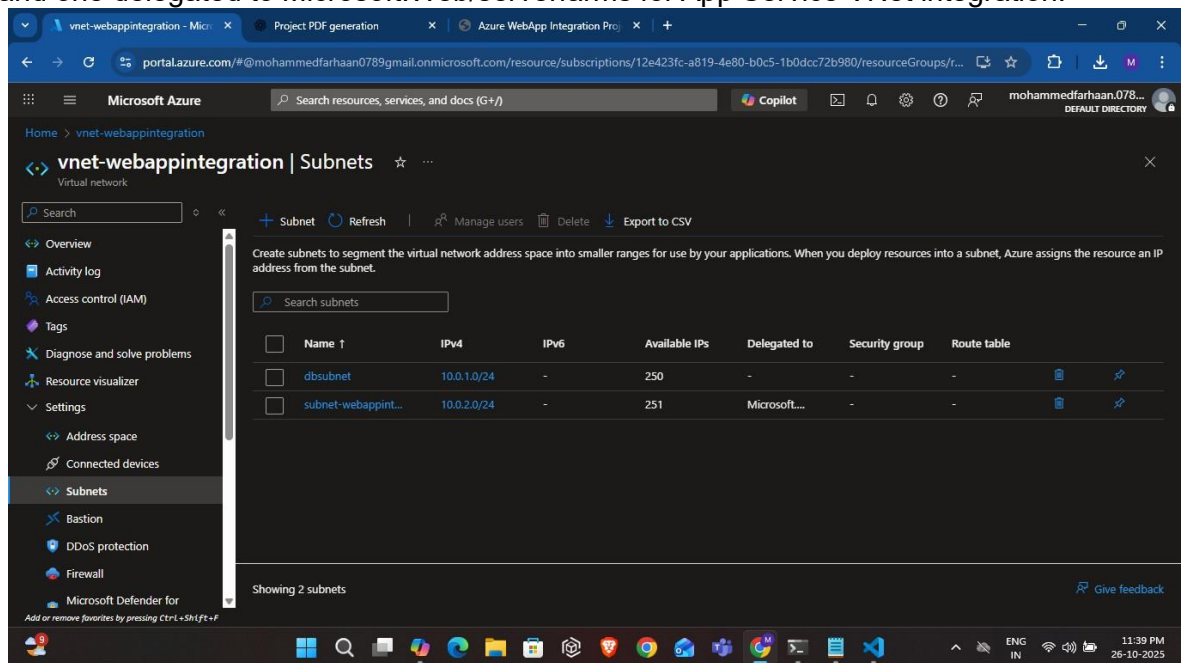
- 1 Created Resource Group **rg-webappintegration** and applied resource lock (CanNotDelete) and tags.



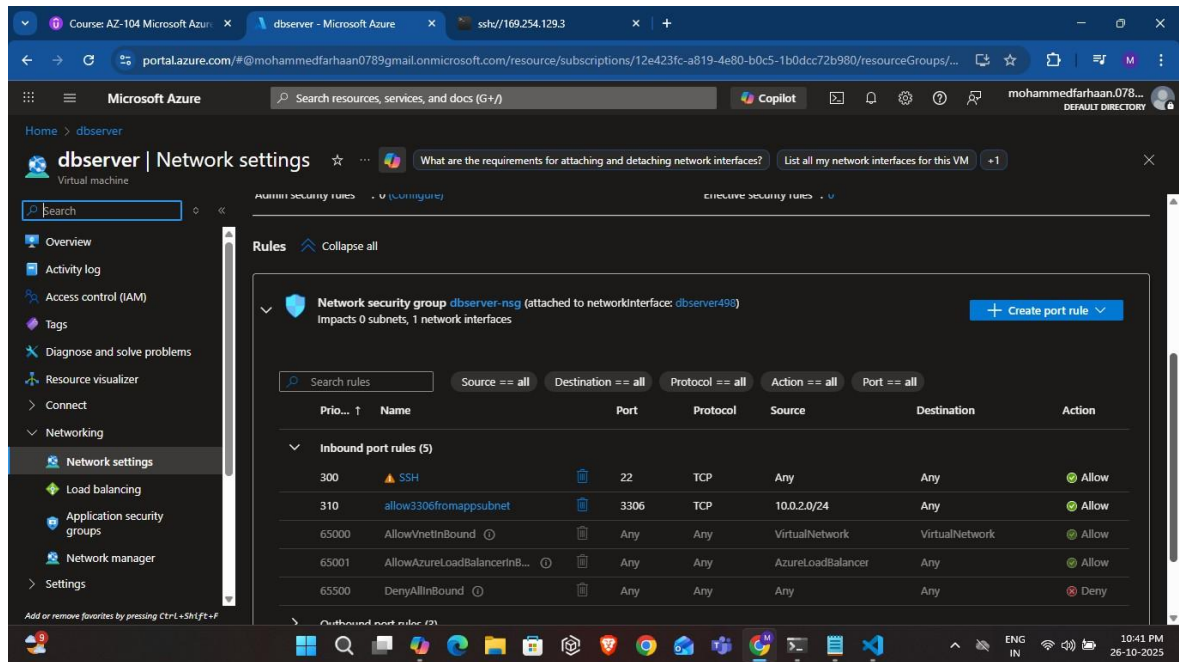
2 Added a new Azure AD user and assigned Reader role at the Resource Group level.



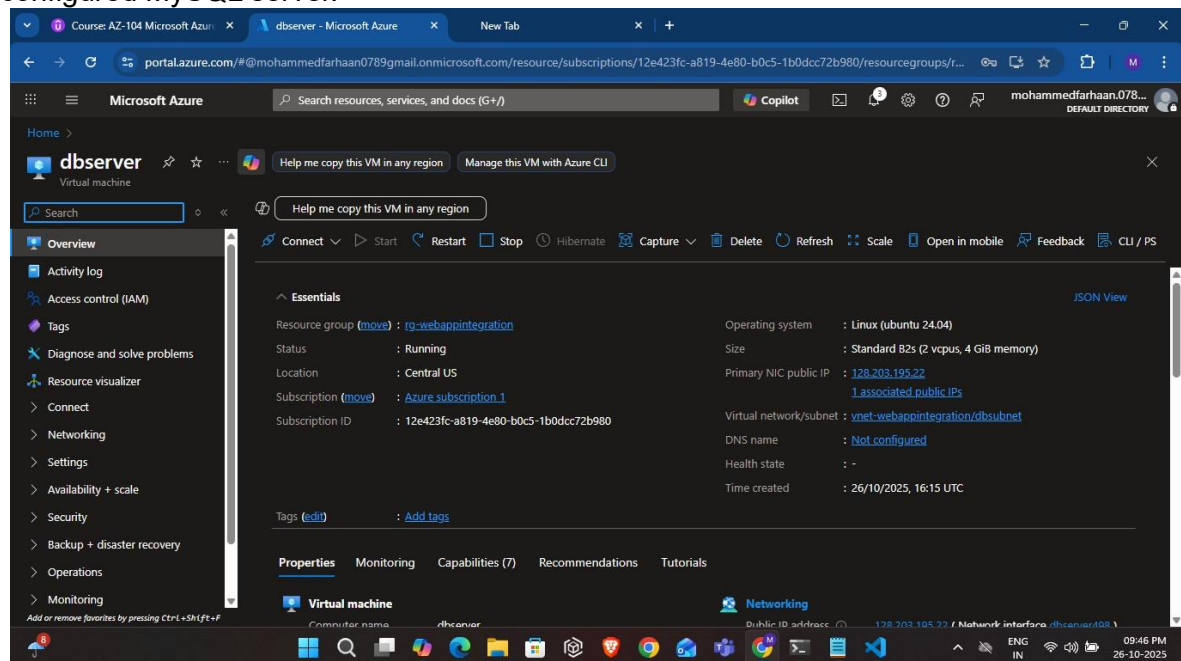
- 3 Deployed Virtual Network **vnet-webappintegration** with two subnets: one for VM and one delegated to Microsoft.Web/serverfarms for App Service VNet integration.



- 4 Created Network Security Group **dbserver-nsg** and allowed inbound MySQL port 3306 only from the App Service subnet.



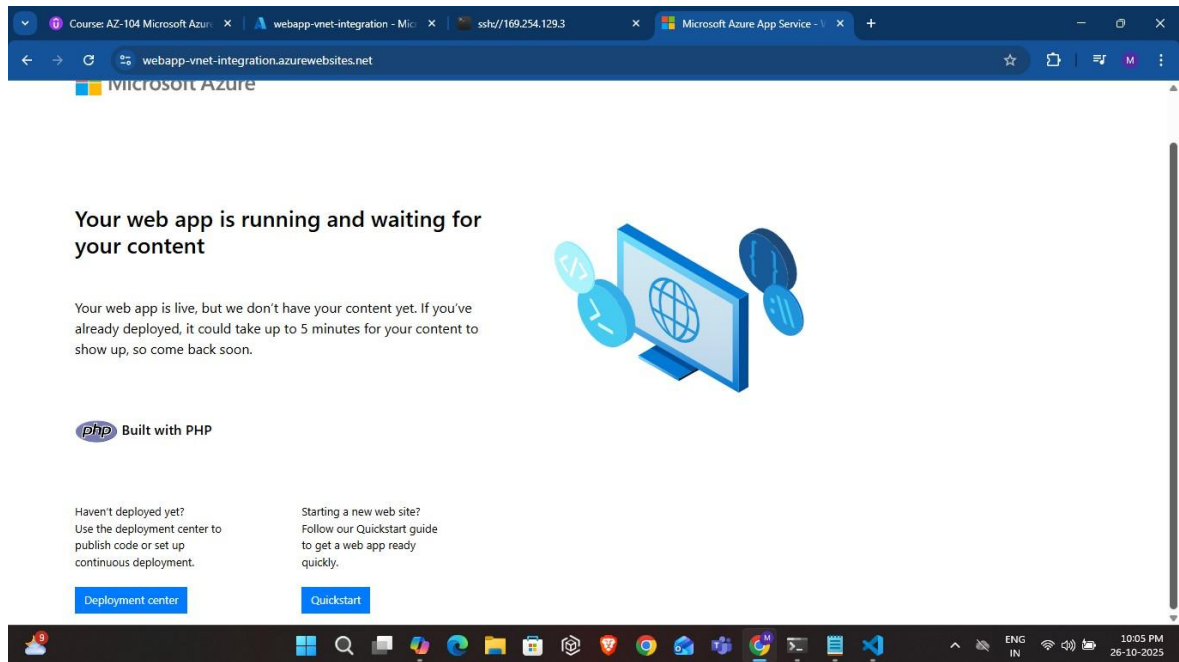
- 5 Deployed Linux Virtual Machine **dbserver** with public IP **dbserver-ip**, and configured MySQL server.



- 6 Created MySQL user, granted privileges, and set bind-address to VM's private IP in MySQL configuration. Inserted sample data into the MySQL table and disassociate public ip of a vm.

• [view commands](#).

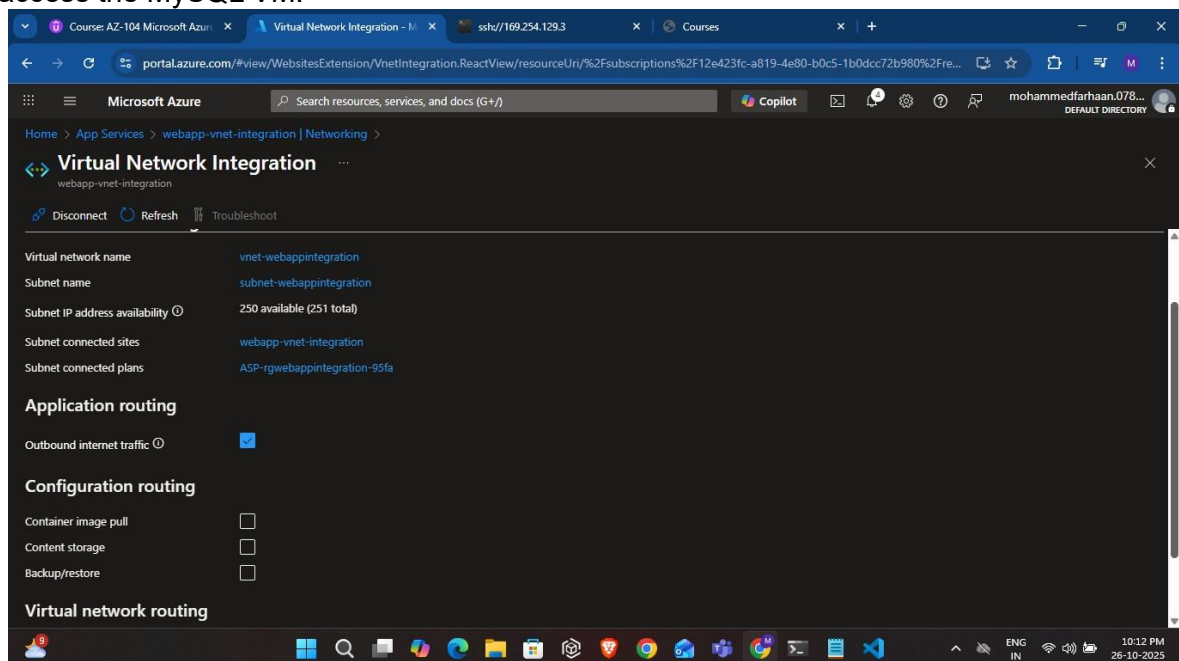
- 7 Deployed App Service Plan **ASP-rgwebappintegration-95fa** and Web App **webapp-vnet-integration** (Linux, PHP).

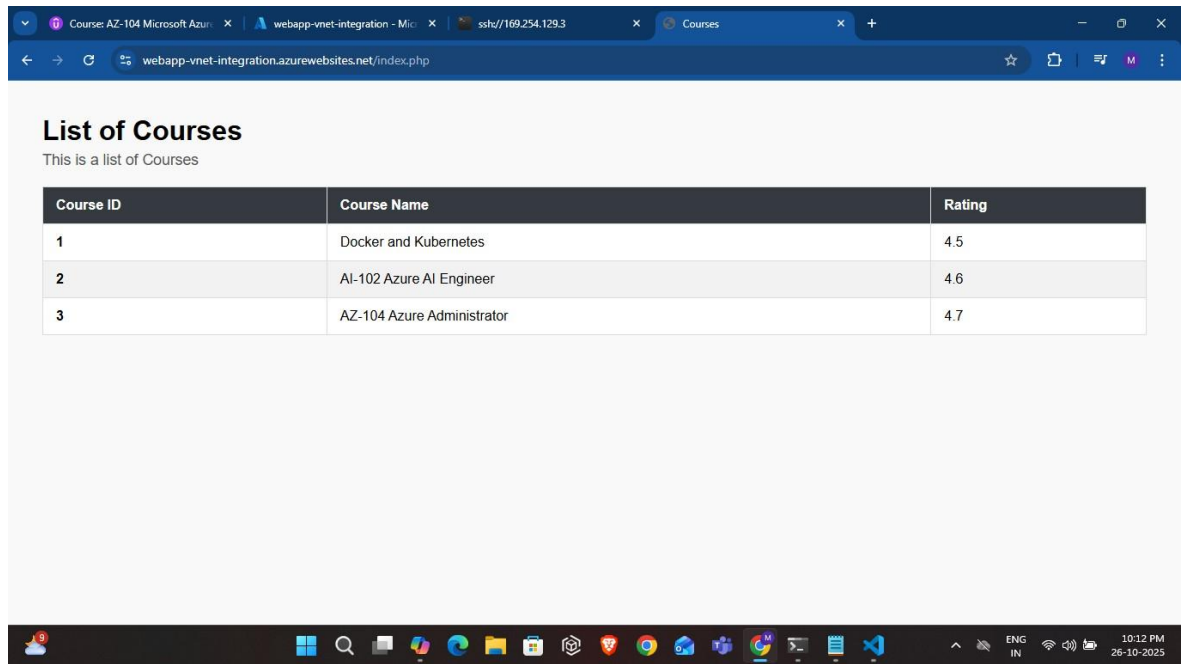


- 8 Manually deployed PHP application via SSH and configured DB connection in code with VM IP, user, DB name, and password.

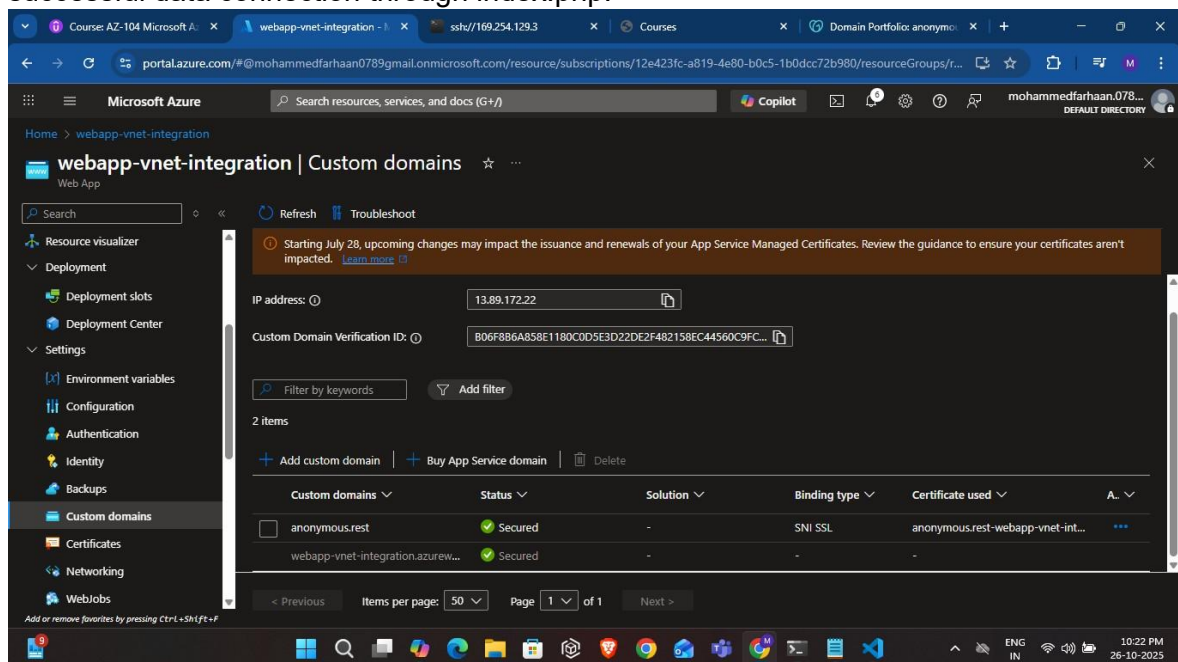
🔗 view source code

- 9 Enabled VNet Integration between App Service and delegated subnet to securely access the MySQL VM.





10 Configured custom domain **anonymous.rest** for the web app and verified successful data connection through index.php.



List of Courses
This is a list of Courses

Course ID	Course Name	Rating
1	Docker and Kubernetes	4.5
2	AI-102 Azure AI Engineer	4.6
3	AZ-104 Azure Administrator	4.7

11 Created Recovery Services Vault ****webappsv**** and enabled VM backup with a manual restore point trigger.

Backup Jobs

Choose columns Filter Export jobs Refresh Feedback

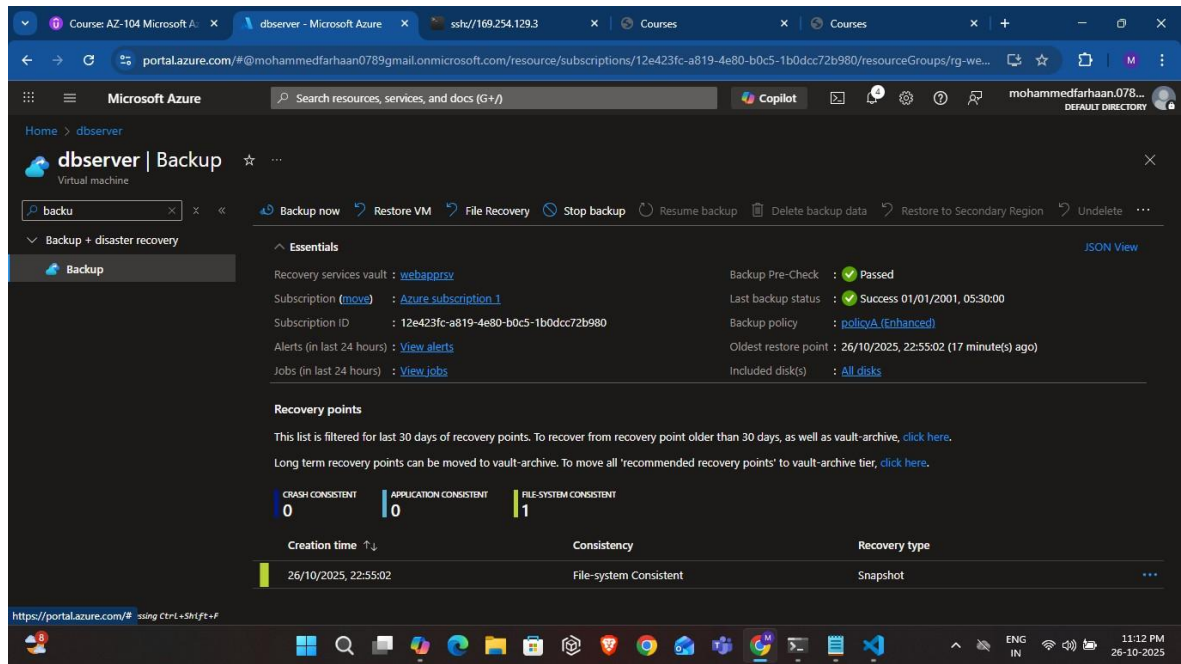
Analyze all job failures for this vault

Filtered by: Item Type - All, Operation - All, Status - All, Start Time - 25/10/2025, 22:55:08, End Time - 26/10/2025, 22:55:08

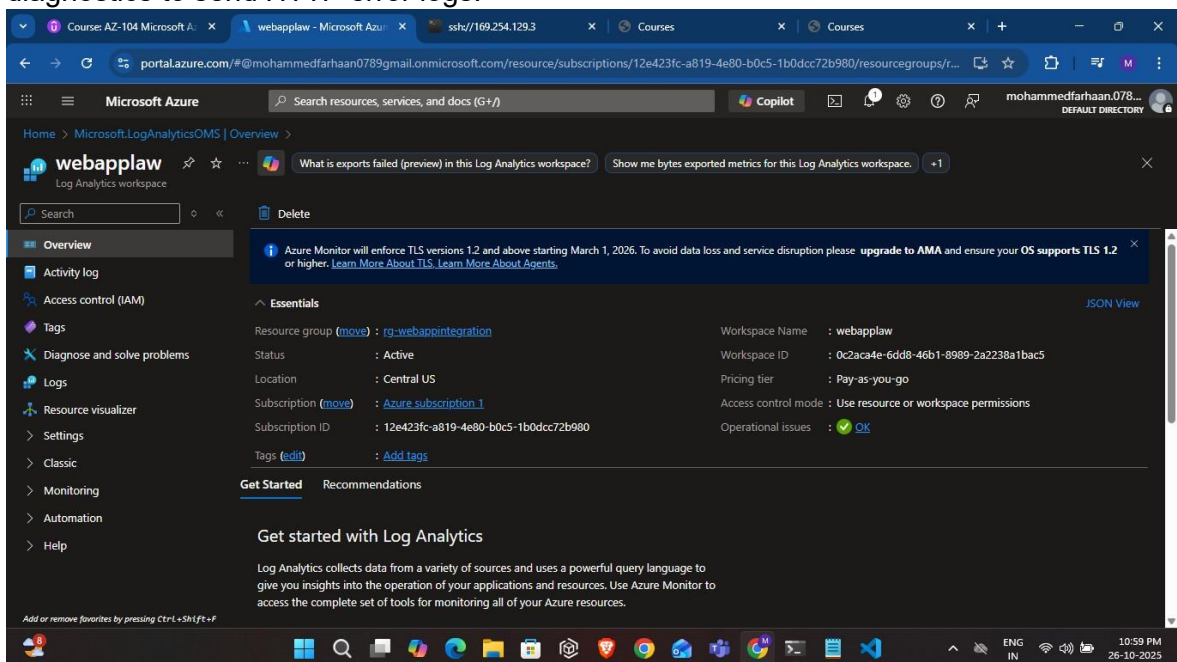
All data fetched from the service.

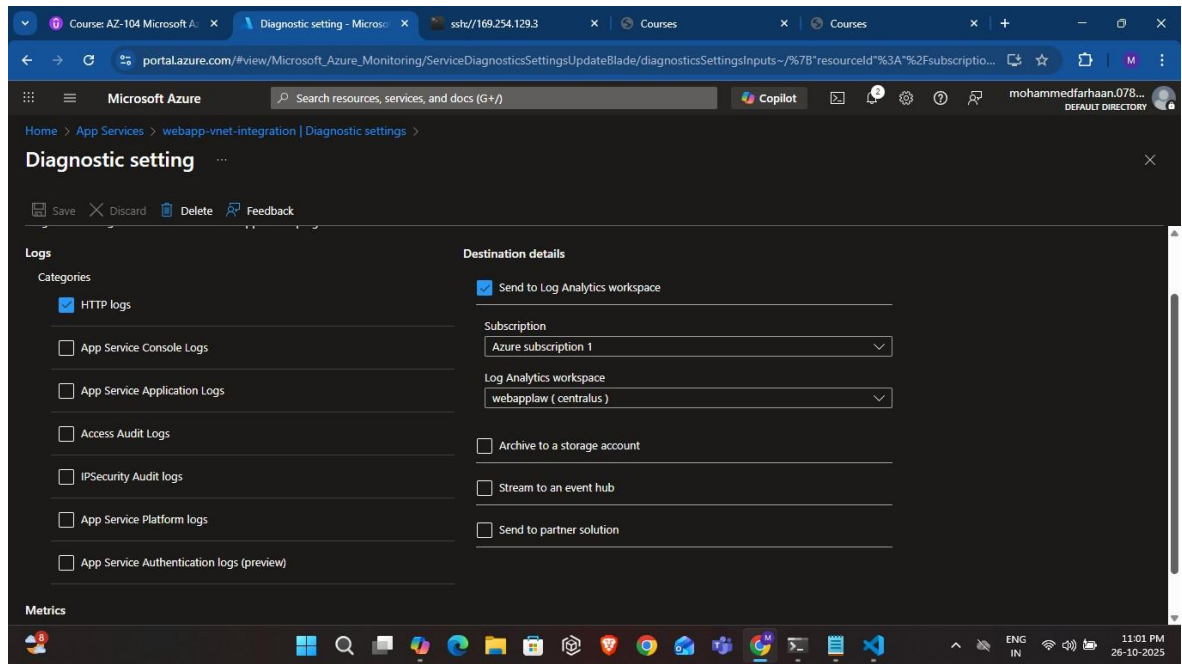
Workload name	Operation	Status	Type	Start time	Total Duration	Details
dbserver	Backup	In progress	Azure Virtual Machine	26/10/2025, 22:54:54	00:00:15	View details
dbserver	Configure backup	Completed	Azure Virtual Machine	26/10/2025, 22:53:21	00:00:30	View details

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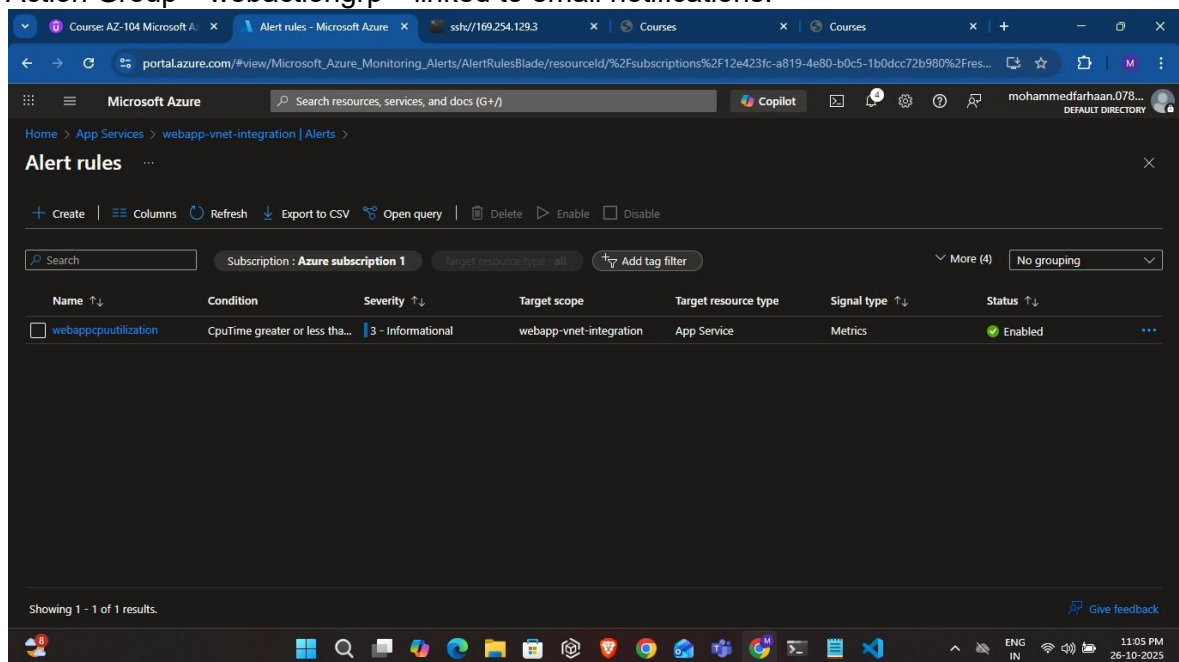


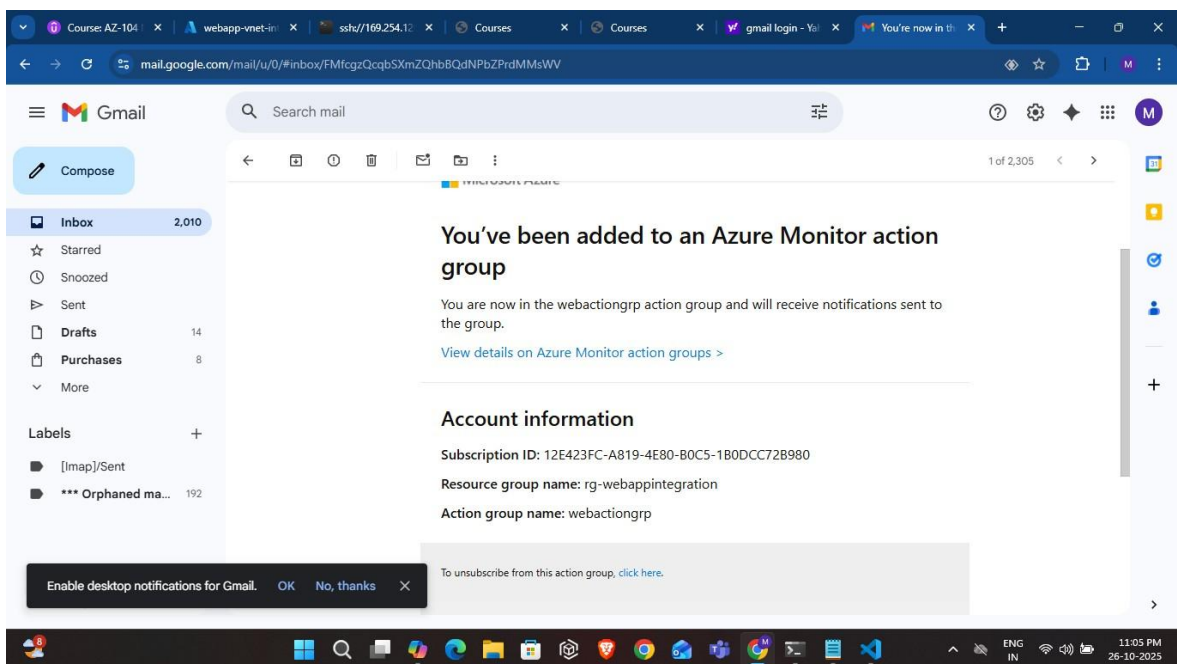
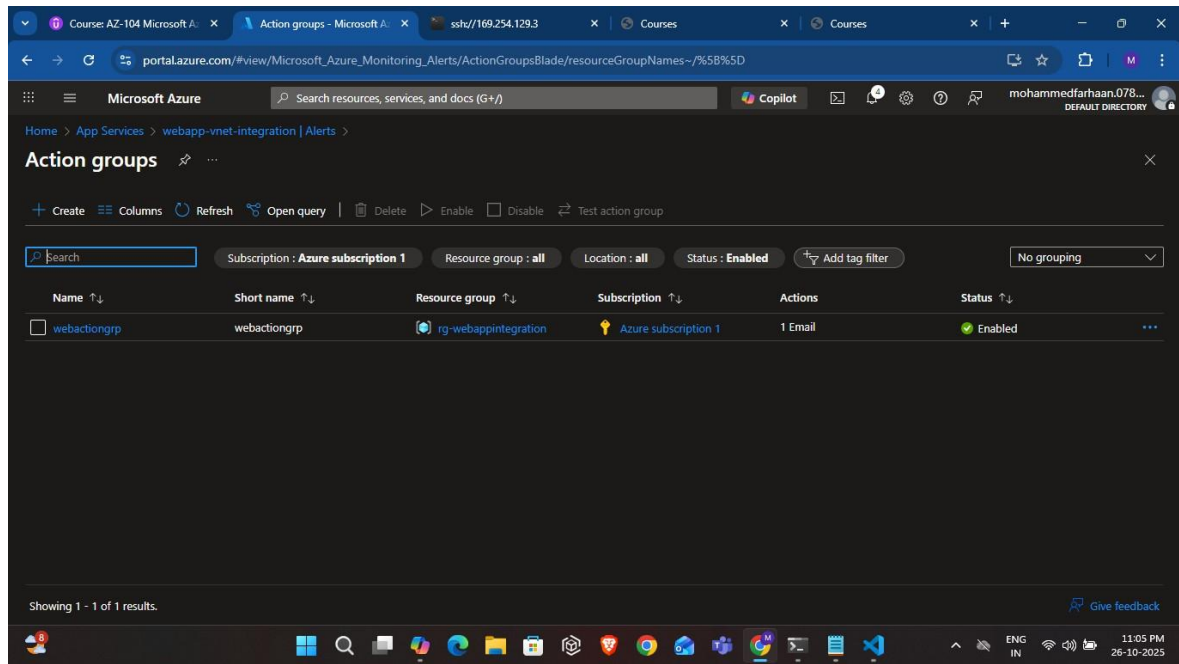
12 Deployed Log Analytics Workspace ****webapplaw**** and connected App Service diagnostics to send HTTP error logs.





13 Configured Alert Rule **webappcpuutilization** with signal metric 'CPU Time' and Action Group **webactiongrp** linked to email notifications.





Outcome

The PHP web application successfully connected to the MySQL database hosted on Azure Linux VM through secure VNet integration. All resources were governed, monitored, and protected with backup and alert configurations.

Skills Demonstrated

- 1 Azure Networking (VNet, NSG, Subnets)

- 2 App Service Deployment & Configuration
- 3 Linux VM & MySQL Database Administration
- 4 Backup & Recovery Configuration using RSV
- 5 Monitoring & Alerts using Log Analytics and Action Groups
- 6 Governance using RBAC, Tags, and Locks

■ Subscription Used

Microsoft Azure Subscription (pay as-you-go)